LuSIR: Latent Upscaling via Self-trained Image Restoration without T2I Pretraining

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Snapshot: masked detail branch v2 and its formal x4 benchmark are complete, Stage 2 clean-fidelity learning-rate probes are complete, the Stage 2 detail-perceptual continuation and shifted-window attention probes have been formally reviewed but not promoted, and several deterministic/generative visible-detail probes were evaluated without producing a clear texture breakthrough. A Stage 1 decoder capacity audit now points the active visible-detail bottleneck back to Stage 2 conditional-latent smoothing. The no-GAN, noise-gated v6 detail branch, Stage 2 latent residual adapter, masked latent residual-shift, and teacher-filtered v7 detail probes are complete; none is promoted. V7 used RealESRGAN only on local highpass patches that were near/better than the base against GT, but step500 still regressed highpass and laplacian detail ratios. A 256-image teacher patch diagnostic then showed that RealESRGAN did not beat the base on PSNR or highpass-L1 for any cached sample, so the v8 probe removed teacher supervision and used GT detail-need masks only during training. V8 stayed stable and selected step 500, but visible detail remained small and step2000 began to lose laplacian ratio, so it is not promoted. The follow-up Stage 2 GT-masked detail probe moved the GT-missing-detail signal into Stage 2 itself via a train-only masked decoded/highpass loss initialized from guarded-detail Stage 2 v2 step 10000, but it was stopped at step1000 after highpass ratio and missing-detail metrics worsened despite a small mean-PSNR increase. A clean-bicubic overfit64 diagnostic is now separated from deployable training: it fixes the first 64 train samples under deterministic bicubic degradation and evaluates on the same set to test whether the current Stage 2 structure/loss can memorize high-frequency detail once generalization is removed. This diagnostic completed 3000 micro-steps and improved train64 from `26.4246` to `29.2477` mean PSNR, `0.7908` to `0.8812` highpass ratio, and `0.01971` to `0.01314` missing-detail energy. A follow-up deterministic512 probe retained the same favorable train-subset trend but failed on held-out clean-bicubic val100: mean PSNR fell by `0.6710 dB`, SSIM by `0.01561`, while highpass error and excess energy increased. Its grids developed repeated ripple/grid texture. The active bottleneck is therefore generalizing a detail-preserving signal without creating high-frequency artifacts, rather than proving basic representational capacity.

`paper/TECHNICAL_REPORT.md` is the canonical report source. `paper/sr_diffusion_report.pdf` and `paper/main.tex` are generated from it with `paper/build_report.sh`, so the Markdown and PDF contain the same report.

1 Objective

LuSIR trains a vision-only x4 latent diffusion super-resolution model without using a pretrained text-to-image backbone. The active task is:

```
LR 128x128 -> HR 512x512
```

The model path is:

```
HR image -> factor-4 VAE -> HR latent
LR image -> condition encoder -> condition latent
noisy HR latent + condition latent + timestep + domain id -> conditional U-Net
denoised latent -> VAE decoder -> SR output
```

The numbered stages describe training order, not a mandatory Stage 1 -> 2 -> 3 -> 4 runtime chain. The current runtime choices are:

```
public Colab default:
  LR -> guarded-detail Stage 2 v2 step 10000 -> Stage 1 decoder

Stage 2 GT-masked detail negative result:
  train: current decoded prediction/GT -> GT missing-detail top20 training mask
  eval:  LR -> GT-masked-detail Stage 2 v3 -> Stage 1 decoder

conservative deterministic option:
  LR -> Stage 2 XL condition encoder -> residual refiner v2 -> Stage 1 decoder
```

```
current detail research candidate:
  LR -> dual-context LSDIR Stage 2 -> Stage 1 decoder
     -> learned detail mask -> masked detail branch v2

teacher-filtered negative result:
  LR -> dual-context LSDIR Stage 2 -> Stage 1 decoder
     -> relaxed learned detail mask -> v7 teacher-filtered detail branch

GT-mask training diagnostic:
  train: base/GT -> GT detail-need top20 training mask
  eval:  LR -> dual-context LSDIR Stage 2 -> Stage 1 decoder
            -> learned detail mask -> v8 detail branch

generative comparison:
  LR -> Stage 2 condition encoder -> Stage 3 OR Stage 4 diffusion U-Net
     -> Stage 1 decoder
```

Stage 4 is a Stage 3-derived replacement diffusion checkpoint. It does not run after Stage 3. Planned Stage 5 distillation would similarly replace the slower diffusion sampler rather than append another serial module.

2 Data

The base photo100k split has 103,450 training images and 100 fixed validation images. The completed dual-context Stage 2 scale-up adds 30,000 unique LSDIR training images, for 133,450 unique training images and the unchanged 100-image validation set. LR inputs are generated on the fly from HR crops. Earlier XL work used `photo_v3_noise_mix`, a strong denoise-focused curriculum with no clean share and 80% combined v2/v3 cases. Later work introduced `photo_detail_mix`: 35% clean, 48% detail-preserving photo degradation, 15% mild degradation, and 2% strong `photo_v2`. This keeps a small robustness tail while shifting the primary objective toward user-facing detail restoration.

3 Completed Baselines

Stage	Run	Result
Stage 1 VAE	`autoencoder_photo10k_b16_eval_online`	`eval/psnr 40.19`
Stage 2 photo100k	`latent_pretrain_photo100k_b64`	best latent loss `0.21230`
Stage 3 photo100k	`diffusion_photo100k_b32`	sampled val100 `25.3745` PSNR
Stage 4 photo100k	`diffusion_photo100k_b32_stage4_condition`	sampled val100 `25.4072` PSNR
Stage 4 photo100k v2	`diffusion_photo100k_b32_stage4_condition_v2`	sampled val100 `22.8426` PSNR, `+0.4323` over bi

The v2 task has a stronger degradation distribution, so its absolute PSNR is not directly comparable with the earlier mild photo100k run.

4 Exploratory Strict-Bicubic Five-Image Comparison

The ordinary LuSIR validation metrics use task-specific degradations, so their absolute PSNR values should not be compared directly with published bicubic-only SR benchmarks. A small strict-bicubic diagnostic was added to separate degradation difficulty from reconstruction capacity. The new `benchmark_bicubic` preset applies only PIL bicubic factor-4 downsampling and does not add blur, noise, compression, color changes, or sharpening.

The shared diagnostic protocol is:

```
source images: DIV2K validation 0801-0805
HR input: deterministic center 512x512 crop
LR input: strict PIL bicubic x4 downsample to 128x128
```

```
metric: mean per-image RGB PSNR over the full 512x512 crop
deterministic inference: CPU FP32
diffusion inference: condition initialization, start timestep 50, 32 DDIM steps
```

Loaded parameter counts include the entire 21.10M-parameter Stage 1 VAE because the current runners load the full module, although inference executes only its 10.52M-parameter decoder. This matches the existing `509.658M` Stage 4 XL accounting.

Inference path	Selected checkpoint	Loaded params	Mean RGB PSNR	vs bicubic
Bicubic	n/a	n/a	29.5999	n/a
Stage 2 XL condition-only	step 72000	40.040M	30.5677	+0.9678
Stage 2 XL + residual refiner v2	step 39000	48.106M	30.8205	+1.2206
Stage 2 multiscale	step 46000	76.591M	31.6068	+2.0069
Stage 2 dual-context LSDIR	step 98000	140.334M	31.7411	+2.1412
Dual-context + detail v1b	step 39500	141.675M	31.8135	+2.2136
Dual-context + detail v1c	step 6000	141.689M	31.8154	+2.2155
Dual-context + detail v1d	step 99500	143.354M	31.9513	+2.3514
Stage 4 XL edge, 32-step sampled	step 4250	509.658M	29.5487	-0.0512

The strict-bicubic result confirms that LuSIR already operates in the 30-32 dB range when the LR input is not additionally corrupted. The much lower `photo_detail_mix` and strong-preset values primarily reflect degradation difficulty rather than a direct 6-7 dB gap to bicubic-only SR papers.

The capacity trend is also informative. Stage 2 XL to multiscale gains `+1.0391 dB`, and multiscale to dual-context gains another `+0.1343 dB`. The detail branches then provide smaller but consistent gains over dual-context: v1b `+0.0725 dB`, v1c `+0.0744 dB`, and the completed 3.02M v1d `+0.2102 dB`. V1d improves v1c by `+0.1358 dB` after three epochs. This confirms that the long capacity run was useful, but the visual change remains conservative and is not a perceptual-detail breakthrough.

The 509.7M Stage 4 XL edge checkpoint is the clearest counterexample to treating parameter count as a quality ranking. On this clean bicubic diagnostic it falls `-0.0512 dB` below bicubic and `-1.0190 dB` below its own Stage 2 XL condition-only path. Visual inspection shows that it can add apparent sharpness, but it also changes clean input details enough to reduce reconstruction fidelity. This is consistent with its strong-noise denoise/color-cleanup training role; it should not be routed unconditionally for clean inputs.

This diagnostic is deliberately small and is not a published-benchmark claim. It uses five center crops, RGB PSNR, no border shave, and PIL bicubic rather than the full-image Y-channel/Matlab-bicubic conventions used by many SR papers. The formal full-image benchmark described next supersedes it for clean-bicubic fidelity comparison.

The machine-readable snapshot is stored in:

`metrics/benchmark_bicubic5_lusir_model_comparison.json`

5 Formal Full-Image x4 Benchmark

A formal evaluator now measures full-image DIV2K validation, Set5, Set14, and Urban100 using their public x4 bicubic LR pairs. It uses MATLAB-compatible BT.601 Y conversion, a four-pixel border shave, and MATLAB-style SSIM. All candidate outputs are required to match the HR dimensions exactly; the evaluator never hides an error by resizing an output.

Candidate	DIV2K	Set5	Set14	Urban100
Bicubic	28.1044	28.4318	26.0928	23.1412
RealESRNet x4plus	28.8250	29.3828	27.1465	24.4613
RealESRGAN x4plus	26.6125	26.6160	25.4216	22.6709
SwinIR classical x4	**31.0838**	not run	not run	not run
LuSIR refiner v2	28.7857	28.1896	27.3704	24.9176
LuSIR dual-context base	29.9575	31.6621	28.2441	25.4816

Candidate	DIV2K	Set5	Set14	Urban100
LuSIR detail v1d	**30.1602**	**31.8892**	**28.4123**	**25.8755**
LuSIR masked detail v2	**30.1636**	**31.9495**	**28.4257**	**25.8922**

The table reports Y PSNR; full Y SSIM and RGB PSNR values are preserved in the machine-readable results. V1d improves the frozen dual-context base on all four datasets. Its Y PSNR gains are `+0.2027`, `+0.2271`, `+0.1682`, and `+0.3939 dB`, respectively. Its Y SSIM is `0.83421`, `0.89440`, `0.77998`, and `0.77875`. This validates the branch redesign under a standard full-image protocol, with the strongest gain on texture-heavy Urban100.

The official SwinIR classical x4 checkpoint reaches `31.0838 / 0.85228` on the same DIV2K evaluator. It is `+0.9235 dB` Y PSNR and `+0.01807` Y SSIM ahead of detail v1d. The detail branch is doing useful corrective work, but this remaining gap identifies the Stage 2/base reconstruction path as the primary clean-fidelity bottleneck.

RealESRNet and RealESRGAN target real-world degradation and perceptual quality, so their clean-bicubic fidelity scores do not establish a general visual quality ranking. Likewise, beating these checkpoints under this protocol does not establish classical-SR SOTA. A classical fidelity baseline, perceptual metrics, real-degradation evaluation, and blind human review remain necessary.

The guarded-detail Stage 2 v2 step-10000 Colab default was also evaluated with test-time augmentation on the same 219-image benchmark using a fast PSNR-only sweep. Off, horizontal flip x2, and full x8 self-ensemble reach `27.8539`, `27.9067`, and `27.9496 dB` mean Y PSNR. Full x8 is therefore a real but small `+0.0957 dB` correction over off. The visual contact sheet shows only subtle differences and the x8 path costs roughly eight times more inference, so TTA is retained as an optional review setting rather than a new default. Masked detail v2 remains higher at `28.1429 dB` in the same PSNR-only comparison.

The Stage 2 latent residual adapter v1 completed the same benchmark after selecting step 11000 on the val100 composite metric. It is not promoted. Against the frozen dual-context Stage 2 base, it trades `-0.0138 dB` mean Y PSNR for `+0.00094` mean Y SSIM. Against the guarded-detail Stage 2 v2 default, it is lower on both mean Y PSNR (`-0.0246 dB`) and mean Y SSIM (`-0.00109`). Against masked detail v2, it trails by `0.3135 dB` Y PSNR and `0.00960` Y SSIM. This is useful negative evidence: a small bounded latent residual can remain stable, but it does not recover visible detail better than the existing masked/detail branch path.

The reproducible protocol and commands are in `docs/SR_BENCHMARK.md`; full machine-readable results are in:

```
metrics/formal_x4_benchmark_lusir_realesr_summary.json
metrics/formal_x4_benchmark_lusir_realesr_metrics.csv
metrics/formal_x4_benchmark_div2k_swinir_summary.json
metrics/formal_x4_benchmark_div2k_swinir_metrics.csv
metrics/formal_x4_benchmark_detail_v2_masked_summary.json
metrics/formal_x4_benchmark_detail_v2_masked_metrics.csv
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_summary.json
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_metrics.csv
metrics/formal_x4_benchmark_stage2_guarded_tta_compare_summary.json
metrics/formal_x4_benchmark_stage2_guarded_tta_compare_metrics.csv
metrics/formal_x4_benchmark_stage2_latent_adapter_v1_value_compare_summary.json
metrics/formal_x4_benchmark_stage2_latent_adapter_v1_value_compare_metrics.csv
```

6 Stage 2 Clean-Fidelity Continuation and Learning-Rate Probes

The clean-bicubic continuation starts from dual-context Stage 2 best98000 and uses `benchmark_bicubic` with a PSNR-oriented decoded loss balance. Its task-specific val100 decoded-PSNR proxy improved gradually:

Step	Decoded PSNR proxy
1000	25.019
4000	25.031
9000	25.045
15000	**25.057**

Step	Decoded PSNR proxy
17000	25.054

The run was stopped manually at step 17825 after no new best beyond step 15000. Its checkpoints remain available for future clean-fidelity work.

These values are not directly comparable with the formal full-image Y-channel results. The formal DIV2K comparison is LuSIR masked detail v2 at `30.1636 dB` and SwinIR classical x4 at `31.0838 dB`.

Separate learning-rate probes preserved the original step-15000 checkpoint. A `20x` continuation collapsed to `15.72 dB` at its first evaluation. A `5x` continuation remained below the original run, while a `5x` from-initialization run reached `25.033 dB` at step 4000 versus `25.031 dB` for the original learning rate. The difference is too small to treat as a gain. The original `5e-6` learning rate is retained, and learning-rate scarcity is rejected as the main bottleneck.

The result also clarifies the objective boundary. Same-objective Stage 2 continuation can refine deterministic fidelity, but it is unlikely to produce the visibly new fine texture expected from a generative model.

7 Stage 2 Deterministic-Subset Capacity Diagnostic

Two clean-bicubic diagnostics removed stochastic degradation and evaluated on the same samples used for training. They are upper-bound capacity tests, not deployable model evaluations.

The first fixed 64-image run completed 3000 micro-steps. Mean PSNR improved from `26.4246` to `29.2477`, highpass energy ratio from `0.7908` to `0.8812`, and missing energy from `0.01971` to `0.01314`. This demonstrates that the current dual-context Stage 2 and Stage 1 decoder can represent substantially more clean detail when generalization is removed.

The follow-up run restarted from dual-context best98000 and expanded the fixed subset to 512 images:

```
config: configs/latent_pretrain_photo130k_lsdir_dual_bicubic_det512_probe.yaml
W&B:   https://wandb.ai/jwheo/LuSIR/runs/0wzx4xzy
steps: 6000
```

On the same train512 subset, mean PSNR improved from `26.95` to `28.40`, highpass ratio from `0.791` to `0.845`, and missing energy from `0.01743` to `0.01410`. However, an additional deterministic clean-bicubic evaluation on held-out val100 reversed the conclusion:

Candidate	Mean PSNR	SSIM	Highpass ratio	Highpass L1	Missing	Excess
Initial best98000	26.9198	0.82143	0.8244	0.03113	0.01799	0.00678
Det512 step6000	26.2488	0.80582	0.8601	0.03366	0.01742	0.00877

The higher held-out highpass ratio is not a quality gain: highpass L1 and excess energy also increase, while PSNR and SSIM regress. Fixed visual grids show repeated ripple/grid texture around foliage, grass, and zebra patterns. The checkpoint therefore memorizes subset-specific high-frequency structure rather than learning a transferable detail prior.

These diagnostics separate capacity from generalization. Further fixed-subset scaling is not prioritized. The next Stage 2 experiment should restore stochastic crops and restrained augmentation, report train and held-out metrics together, and use highpass error/excess energy plus visual artifact checks as guardrails rather than optimizing highpass energy ratio alone.

The follow-up configuration is `configs/latent_pretrain_photo130k_lsdir_dual_bicubic_generalization_v1.yaml`. It restarts from best98000, restores full-split stochastic crops, horizontal flips, texture-aware crop selection, and restrained color jitter, while using held-out deterministic val100 as the primary evaluation and checkpoint selection set. A fixed deterministic train512 set is evaluated in parallel to expose train/validation divergence. The trainer now also supports an `artifact_excess` soft-hinge loss that penalizes local predicted highpass energy above the GT evidence. The 12000-step probe uses batch 8 without gradient accumulation and retains only best/latest checkpoints. Promotion requires held-out PSNR/SSIM stability together with non-increasing highpass error and excess energy.

V1 was stopped after the first trained evaluation. At step500, held-out mean PSNR improved by `0.0661 dB` and excess energy decreased from `0.00678` to `0.00555`, but SSIM fell by `0.00265`, highpass ratio fell from `0.82445` to `0.77892`, and missing energy increased from `0.01799` to `0.01935`. This is a smoother distortion-oriented solution, not a detail-generalization success.

V2 keeps the data, model, learning rate, and dual evaluation unchanged. It reduces the excess-hinge weight from `2.0` to `1.0` and adds a target-aligned missing-detail hinge at weight `2.0`. The missing term projects predicted highpass onto the GT highpass sign, providing a non-zero, phase-correct gradient even when the current prediction is smooth. The excess term still uses absolute local energy, preserving a separate guard against unsupported texture.

V2 was stopped after step1000. SSIM increased from `0.82145` to `0.82921` and missing energy decreased from `0.01798` to `0.01533`, confirming that the target-aligned gradient restores structural high-frequency response. The weight was too strong: highpass ratio increased from `0.82468` to `0.90093`, excess energy from `0.00679` to `0.00915`, and mean PSNR fell by `0.0510 dB`.

V3 isolates the balance rather than changing architecture or data again. It uses excess weight `1.5` and missing weight `0.8`, interpolated between the V1 under-detail and V2 over-detail boundaries. All other training and evaluation settings remain fixed.

V3 completed 12000 steps and selected step11500 by held-out clean-bicubic val100 mean PSNR. Relative to the initial dual-context checkpoint, mean PSNR increases from `26.91987` to `27.05483 dB`, SSIM from `0.82145` to `0.82665`, highpass L1 decreases from `0.031130` to `0.030733`, and missing energy decreases from `0.017983` to `0.017283`. The selected grid shows no obvious ripple pattern or oversharpening.

The gain transfers to the formal 219-image clean-bicubic benchmark:

Candidate	Mean Y PSNR	Mean Y SSIM	Mean RGB PSNR	Mean RGB SSIM
Dual-context step98000	27.84314	0.79742	26.31306	0.77340
Generalization V3 step11500	**27.99167**	**0.80295**	**26.47050**	**0.77969**

This result does not transfer to real-degradation robustness. Relative to step98000, decoded PSNR regresses by `0.2125 dB` on `mild`, `0.2749 dB` on `photo_detail_mix`, `0.9644 dB` on `photo_v2`, and `0.7808 dB` on `photo_v3_noise_mix`. V3 is therefore preserved as the strongest standalone clean-bicubic Stage 2 research checkpoint, but it does not replace the public HF or Colab default. A follow-up must mix clean and progressively stronger degradations while retaining both formal clean metrics and all four cross-preset evaluations as guardrails.

To test whether residual clean-fidelity error is capacity limited, the next probe expands only the full-resolution residual trunk from 16 to 40 blocks. This raises Stage 2 capacity from `119.238M` to `147.587M` parameters (`1.238x`) while leaving both multiscale context branches unchanged. The new blocks use zero-initialized second convolutions, so partial initialization from V3 step11500 reproduces the original output exactly before training. The 4000-step probe evaluates clean bicubic plus `mild`, `photo_detail_mix`, `photo_v2`, and `photo_v3_noise_mix` every 500 steps. On one L40S, batch 6 without gradient accumulation is the largest stable configuration across training and all evaluations, using approximately `39.2/46.1GB` VRAM. A clean gain below `0.05 dB`, or renewed degradation-preset regression, is evidence that capacity is not the primary bottleneck.

The probe completed 4000 steps and selected step3500. Held-out clean-bicubic val100 mean PSNR improved by only `0.00364 dB`. All robustness evaluations regressed: `-0.02011 dB` on `mild`, `-0.01005 dB` on `photo_detail_mix`, `-0.05466 dB` on `photo_v2`, and `-0.06268 dB` on `photo_v3_noise_mix`.

The formal 219-image benchmark confirms that the small clean gain is not practically meaningful:

Candidate	Mean Y PSNR	Mean Y SSIM	Mean RGB PSNR	Mean RGB SSIM
Generalization V3 step11500	27.99167	0.802953	26.47050	0.779686
148M trunk step3500	**27.99716**	**0.803327**	**26.47654**	**0.780054**

The Y-PSNR gain is `0.00549 dB` with only `114/219` per-image wins. DIV2K improves by `0.00189 dB` while winning only `44/100` images. Fixed grids and the formal contact sheet show no distinguishable visual improvement. The additional 28.35M parameters and higher memory cost are therefore not justified. The result rejects simple Stage 2 trunk capacity as the primary bottleneck and redirects the next experiment toward a mixed clean/real degradation curriculum.

8 Stage 2 Detail-Perceptual Continuation Review

A separate Stage 2 continuation started from dual-context best98000 and trained on `photo_detail_mix` with lower latent/decoded weight, stronger highpass magnitude, VGG feature loss, and a detail-score selection metric:

```
config: configs/latent_pretrain_photo130k_lsdir_dual_detail_perceptual_v1.yaml
run:    /home/ubuntu/scratch/sr-diffusion/runs/latent_pretrain_photo130k_lsdir_dual_detail_perceptual_v1
W&B:    https://wandb.ai/jwheo/LuSIR/runs/hybqq4rj
```

The val100 proxy confirmed that the loss can raise high-frequency energy: step6000 reached decoded PSNR `24.5921` with Laplacian energy ratio `0.3824`, and latest step12000 reached decoded PSNR `24.6144` with ratio `0.3533`. The original dual-context step98000 baseline on the same proxy was `24.6197` and `0.3123`.

The decisive review used the same 219-image formal x4 benchmark as above, but with the Stage 2 base path only:

Candidate	Mean Y PSNR	Mean Y SSIM	Mean RGB PSNR	Mean RGB SSIM
Bicubic	25.7170	0.71773	24.2697	0.69205
Dual-context step98000	**27.8431**	0.79742	**26.3131**	0.77340
Detail-perceptual step6000	27.7737	**0.79914**	26.2482	**0.77506**
Detail-perceptual latest12000	27.8356	0.79827	26.3107	0.77417

Relative to dual-context step98000, step6000 loses `0.0694 dB` Y PSNR while gaining `0.00172` Y SSIM. Latest step12000 is almost tied: `-0.0076 dB` Y PSNR, `+0.00085` Y SSIM, and `156/219` Y-SSIM wins. Dataset-level latest12000 deltas are `-0.0193 dB` on DIV2K, `+0.0272 dB` on Set5, `-0.0555 dB` on Set14, and `+0.0092 dB` on Urban100.

Visual crop review shows small improvements on some building/window grid regions, but not a clear user-visible texture breakthrough. The conclusion is therefore conservative: keep dual-context step98000 as the default Stage 2 checkpoint, preserve latest12000 as an optional SSIM/detail-biased research candidate, and avoid spending the next run on a longer continuation of the same objective.

The new machine-readable results and visual crop sheet are:

```
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_summary.json
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_metrics.csv
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_delta_crop_selection.csv
samples/stage2_detail_perceptual_v1_benchmark_delta_crop_sheet.jpg
```

9 Stage 2 Shifted-Window Attention Probe Review

The next Stage 2/base architecture probe added shifted-window self-attention and a gated depthwise feed-forward branch to the dual-context predictor. The goal was to test whether broader feature mixing could reduce conditional-mean smoothing in repetitive structures such as windows, grids, leaves, and small texture.

Three variants were evaluated on the same val100 proxy:

Candidate	Stop step	Decoded PSNR range	Detail-ratio range	Best score
v2, implementation bug	8000	24.60-24.63	0.315-0.343	24.950
v3 true-dual, window 8	6000	24.60-24.63	0.315-0.343	24.953
v3 true-dual, window 12	4000	24.60-24.63	0.315-0.341	24.954

The v2 probe is retained only as a cautionary result: the `dual_multiscale_attention` implementation initially did not enable the second `extra_context` branch, so it compared single-context-plus-attention against the dual-context baseline. V3 corrected this by preserving the full dual-context path and applying attention after `context + extra_context`. Parameter groups then trained the new attention branch at `2e-5` while keeping the inherited trunk/context/output at `5e-6`, with a warmup-cosine scheduler.

The corrected `8x8` and `12x12` probes both stayed near the dual-context baseline and did not produce a visible texture breakthrough. Increasing window size raised memory and compute cost (`12x12` used about `31.5 / 46.1 GB` on a single L40S and ran at about `1.83` micro-steps/s) without changing the trajectory. The interpretation is that attention window size is not the primary bottleneck. The model still learns a conservative deterministic latent estimate under the current supervision. Future Stage 2 work should target a residual/detail correction path on top of a preserved fidelity base, or decoder-side detail capacity, rather than continuing window-size scaling.

10 Stage 1 Decoder Capacity Audit

After the masked v5 PatchGAN detail-branch probe collapsed fidelity inside the selected detail mask, the next diagnostic separated Stage 1 decoder capacity from Stage 2 conditional-latent smoothing. A new tool, `tools/analysis/audit_stage1_decoder_detail_capacity.py`, evaluates HR autoencoding through Stage 1 and compares it with the Stage 2 decoded base on the same `photo_detail_mix` val100 set.

Path	Mean PSNR	SSIM	Highpass ratio	Laplacian ratio	Missing energy
Stage 1 VAE recon	41.8121	0.99187	0.9965	0.9553	0.00174
Stage 2 dual-context base	26.4889	0.80013	0.7886	0.3191	0.01968

The result weakens the hypothesis that the current Stage 1 decoder is the main visible-detail bottleneck. When given the HR latent, it preserves nearly all highpass energy and most Laplacian energy. The same decoder fed by Stage 2 dual-context best98000 loses far more high-frequency structure. The active bottleneck is therefore Stage 2 conditional-mean smoothing, not decoder capacity.

This audit also explains a recurring metric mismatch. The legacy Stage 2 training eval reports `eval/decoded_psnr` from global MSE, whereas the audit reports mean per-image PSNR. For the same dual-context checkpoint on the same val100 set these are `24.6197` and `26.4889`, respectively. The Stage 2 trainer now logs both views, plus `eval/decoded_ssim`, `eval/highpass_energy_ratio`, `eval/missing_energy`, `eval/excess_energy`, and `eval/mean_psnr_detail_score`.

The guarded follow-up configuration is `configs/latent_pretrain_photo130k_lsdir_dual_detail_guarded_v2.yaml`. It starts from dual-context best98000, keeps the architecture unchanged, avoids VGG/GAN pressure, and slightly strengthens decoded highpass supervision. Its selection metric, `decoded_mean_psnr + 2 * highpass_energy_ratio`, is only a shortlist metric; visual grids, formal benchmark scores, and artifact review remain the promotion criteria.

The first separate generative probe is `configs/diffusion_photo130k_lsdir_highfreq_residual_v1_b8.yaml`. It freezes the dual-context Stage 2 base and Stage 1 decoder, then trains a 76.6M gated/bounded residual U-Net with strong high-frequency supervision and a lowpass anchor. The output layer is zero-initialized: the smoke-test prediction and condition images were byte-identical, with zero latent residual and zero decoded drift before the first optimizer update.

11 Stage 2 XL Candidate Selection

The XL condition encoder uses:

```
config: configs/latent_pretrain_photo100k_v3_noise_xl.yaml
base_channels: 256
num_blocks: 16
degradation: photo_v3_noise_mix
finished step: 80000
```

Candidate comparison on the same validation images:

Candidate	Step	Latent loss	Latent MSE	Decoded PSNR
best_eval_latent	66000	0.27230	0.91770	21.3828
step_0072000	72000	0.27940	0.97295	21.5241
latest	80000	0.27593	0.89609	21.5062

The 72k checkpoint was selected for Stage 4 XL because it had the best decoded condition-only PSNR on the fixed validation set.

12 Stage 4 XL Edge-Loss Result

The first XL diffusion run used the selected Stage 2 XL condition encoder and partial initialization from the smaller Stage 4 v2 checkpoint.

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_edge_b16
U-Net path: 469.6M parameters
full inference path: 509.658M parameters
train batch size: 16
GPUs: 2x A100-SXM4-80GB through PyTorch DDP
finished step: 5000
selected checkpoint: step 4250, best eval/decoded_mse
```

Training-time best proxy:

```
step 4250
eval/decoded_psnr: 21.9872
eval/decoded_mse: 0.025313
eval/noise_mse: 27.66008
eval/x0_mse: 0.86226
```

Sampled validation evaluation:

```
checkpoint: best_eval_condition_decoded.pt
checkpoint step: 4250
split: val
limit: 100
init: condition
start_timestep: 50
steps: 32
mean_bicubic_psnr: 22.3599
mean_sr_psnr: 23.0793
mean_psnr_delta: +0.7195
```

The latest XL Stage 4 checkpoint is therefore better than bicubic on the current v3 validation setup and better aligned with the desired denoise/color cleanup behavior than the Stage 2 condition-only path. It is still not a final restoration model: outputs remain softer than GT on fine textures.

13 Stage 4 XL Role-Split and Gated-Residual Probes

Follow-up probes tested whether Stage 4 could act as a bounded detail refiner instead of freely overwriting the deterministic Stage 2 condition latent.

The role-split mild probe added a lowpass anchor and separated low/high-frequency decoded losses:

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_rolesplit_mild_b8_probe.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_rolesplit_mild_b8_probe
W&B: https://wandb.ai/jwheo/LuSIR/runs/lrb6nco9
```

```
finished step: 8000 micro-steps, 2000 optimizer updates
best one-step checkpoint: step 7500, decoded PSNR 23.2515
```

Sampled mild val100 showed that role-split losses reduced some over-editing at very low start timesteps, but the diffusion path still damaged the Stage 2 condition output on average:

Model	Start timestep	Mean SR PSNR	vs condition-only	Wins vs condition
Stage 2 XL condition-only	n/a	25.0449	n/a	n/a
Stage 4 role-split	25	24.5747	-0.4702	3/100
Stage 4 role-split	10	24.9185	-0.1264	3/100
Stage 4 role-split	5	24.9935	-0.0514	6/100
Stage 4 role-split	1	25.0335	-0.0114	10/100

The next probe changed the diffusion prediction parameterization to `gated_residual_x0`. Instead of predicting the full denoised latent, the U-Net predicts a bounded residual and gate on top of the condition latent:

```
x0_hat = condition + residual_scale * tanh(residual_logits) * sigmoid(gate_logits + gate_bias)
residual_scale: 1.25
gate_bias: 0.0
out_channels: 32
```

The probe was initialized from the role-split best checkpoint with partial initialization and stopped at step 2000 after the sampled validation result reached condition-only parity:

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_gated_residual_mild_b8_probe.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_gated_residual_mild_b8_probe
W&B: https://wandb.ai/jwheo/LuSIR/runs/edfko8e8
best one-step checkpoint: step 1000
final checkpoint: step 2000
```

Sampled mild val100:

Model	Checkpoint	Start timestep	Mean SR PSNR	vs condition-only	Wins vs condition
Stage 2 XL condition-only	n/a	n/a	25.0449	n/a	n/a
Stage 4 gated residual	1000	25	25.0415	-0.0035	25/100
Stage 4 gated residual	1000	10	25.0415	-0.0034	25/100
Stage 4 gated residual	1000	5	25.0416	-0.0034	25/100
Stage 4 gated residual	1000	1	25.0418	-0.0032	25/100
Stage 4 gated residual	2000	25	25.0445	-0.0004	34/100
Stage 4 gated residual	2000	10	25.0444	-0.0006	32/100
Stage 4 gated residual	2000	5	25.0444	-0.0005	31/100
Stage 4 gated residual	2000	1	25.0443	-0.0007	32/100

This is a partial success. The gated residual parameterization almost eliminated the condition damage seen in previous Stage 4 probes and increased the number of samples beating condition-only from `10/100` at role-split t1 to `34/100` at gated-residual step2000 t25. However, the mean result still does not beat the Stage 2 condition-only baseline. The current interpretation is that Stage 4 has learned a safe near-identity residual path, not a reliable missing-detail generator.

14 Stage 2 Residual Diagnostic and Deterministic Refiner

A direct residual/oracle diagnostic was run to separate the Stage 2 condition-only error into lowpass and highpass components on mild val100:

Metric	Value
Bicubic PSNR	24.4778
Condition decoded PSNR	25.0543
Oracle full residual PSNR	41.8207
Oracle full vs condition	+16.7664
Oracle highpass PSNR	35.0872
Oracle highpass vs condition	+10.0329
Oracle lowpass PSNR	25.0814
Oracle lowpass vs condition	+0.0270
Residual highpass energy ratio	0.8988
Residual lowpass energy ratio	0.0758

This indicates that the Stage 2 condition encoder already preserves most structure and low-frequency color, while the remaining recoverable error is dominated by high-frequency detail.

The follow-up deterministic bounded residual refiner freezes the Stage 1 VAE and Stage 2 condition encoder, then predicts only:

```
condition + residual_scale * tanh(residual_logits) * sigmoid(gate_logits + gate_bias)
```

The sparse-gate probe reached its best validation result at step 500:

Model	Mean PSNR	vs condition	Wins vs condition	Gate mean
Stage 2 condition-only	25.0449	n/a	n/a	n/a
Sparse-gate residual refiner	25.1178	+0.0729	86/100	0.2147
Open-gate residual refiner	25.0972	+0.0523	73/100	0.8680

The sparse-gate result is a small but real gain and is qualitatively close to the condition output, without the destructive edits seen in previous diffusion Stage 4 probes. The open-gate ablation was worse despite a much larger mean gate value, so the next step should not simply force larger residual edits.

The sparse-gate refiner was then connected to standalone eval and inference scripts and evaluated without retraining across the active degradation presets:

Degradation	Bicubic PSNR	Condition PSNR	Refined PSNR	vs condition	Wins
`mild`	24.4778	25.0449	25.1178	+0.0729	86/100
`photo_v2`	22.4103	22.9271	22.9767	+0.0496	77/100
`photo_v3_noise_mix`	22.3599	22.9014	22.9600	+0.0586	86/100

Qualitatively, the refined outputs remain very close to the Stage 2 condition outputs. This is useful because the refiner does not introduce the destructive edits seen in earlier diffusion probes, but the visible detail gain is still small. The result is best interpreted as a safe residual teacher or warm start, not as a final detail generator.

15 Teacher-Supervised Stage 4 Residual Probe

The sparse-gate deterministic refiner was used as a frozen teacher for the gated-residual Stage 4 U-Net. Direct losses supervised the predicted residual, highpass residual, and gate on `photo_v3_noise_mix`.

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_v3_b8_probe.yaml
train batch size: 8
gradient accumulation: 4
finished step: 8000 micro-steps, 2000 optimizer updates
```

Sampled val100, condition initialization, 32 sampling steps:

Checkpoint	Start timestep	Mean SR PSNR	vs bicubic	vs condition	Wins vs condition
Teacher Stage 4 step 2000	25	22.9640	+0.6041	+0.0626	68/100
Teacher Stage 4 step 2000	50	22.9639	+0.6040	+0.0625	n/a
Teacher Stage 4 step 4000	25	22.9571	+0.5972	+0.0557	65/100
Teacher Stage 4 step 8000	25	22.9490	+0.5891	+0.0476	59/100
Existing edge Stage 4 step 4250	25	22.9563	+0.5964	+0.0549	42/100
Existing edge Stage 4 step 4250	50	23.0799	+0.7200	+0.1784	45/100

Teacher supervision therefore produced a small, stable cleanup gain over the Stage 2 condition output and slightly exceeded the edge model at t25. However, the user-facing objective was not achieved. Fur, leaves, branches, and building detail remained strongly smoothed. A simple absolute-Laplacian diagnostic measured teacher step-2000 output at 21.8% of GT detail energy, below the existing edge t25 output at 32.7%. This is not a complete perceptual metric, but it agrees with visual inspection.

The best sampled checkpoint was step 2000; continuing to step 8000 reduced both mean PSNR and condition win count. The active interpretation is that the current `photo_v3_noise_mix` curriculum overemphasizes severe denoise/cleanup cases, so direct residual supervision teaches a safer cleanup operator rather than a missing-detail generator.

16 Detail-Preserving Curriculum Adaptation

A fixed-sample degradation audit confirmed the curriculum mismatch. On val100, `photo_v3_noise_mix` reduced the bicubic baseline to 22.3599 PSNR and produced mean LR chroma RMS error 0.02040 relative to clean downsampling. `photo_detail_mix` increased the bicubic baseline to 24.7357 and reduced the chroma error to 0.00507.

The existing Stage 2 XL condition encoder was evaluated before retraining. It already improved `photo_detail_mix` from 24.7357 bicubic PSNR to 25.3103, a +0.5745 gain. Stage 2 was therefore frozen, and the teacher-supervised gated-residual Stage 4 was adapted from its selected step-2000 checkpoint:

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_detail_b8_long.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_detail_b8_long
W&B: https://wandb.ai/jwheo/LuSIR/runs/so0lbyte
train batch size: 8
gradient accumulation: 4
learning rate: 1e-6
finished step: 12000 micro-steps, 3000 optimizer updates
```

Sampled `photo_detail_mix` val100, condition initialization, start timestep 25, 32 sampling steps:

Model/checkpoint	Mean SR PSNR	vs bicubic	vs condition	Wins vs condition
Stage 2 condition-only	25.3103	+0.5745	n/a	n/a
Teacher Stage 4 initialization	25.3187	+0.5829	+0.0084	46/100
Photo-detail Stage 4 step 8000	25.3406	+0.6049	+0.0303	71/100
Photo-detail Stage 4 step 12000	25.3337	+0.5980	+0.0235	67/100
Existing edge Stage 4 step 4250	25.1176	+0.3818	-0.1927	13/100

Step 8000 is the selected checkpoint. This is the first sampled gated-residual Stage 4 result that beats the Stage 2 condition-only baseline on both mean PSNR and a clear majority of samples. Qualitatively, it preserves the condition structure and sharpness and avoids the broad destructive edits produced by the edge model on this distribution.

The result remains conservative. Mean absolute-Laplacian energy is 29.7% of GT for step 8000, compared with 29.6% for its teacher initialization and 41.2% for the more aggressive edge model. The gain therefore comes primarily from more accurate bounded corrections, not from strong new texture synthesis. Rare strong-tail samples can still exhibit bright artifacts.

17 Decoded-Detail Residual Refiner v2

The deterministic bounded residual refiner uses 192 hidden channels, 12 residual blocks, and decoded-image/highpass supervision through the frozen VAE decoder. An initial 12000-step run was continued to 40000 micro-steps with a lower `2.5e-5` learning rate. Step 39000 achieved the best global decoded PSNR and was selected.

Degradation	Condition mean PSNR	Refined mean PSNR	Gain	Wins
`photo_detail_mix`	25.3103	25.6410	+0.3307	94/100
`mild`	25.0449	25.3161	+0.2712	91/100
`photo_v2`	22.9271	23.0419	+0.1148	81/100
`photo_v3_noise_mix`	22.9014	23.0787	+0.1773	81/100

On the training curriculum, step 39000 reached global decoded PSNR `24.0305`, `+0.2927 dB` over condition-only, mean PSNR gain `+0.3307 dB`, SSIM gain `+0.01076`, and wins on `94/100` images. Step 40000 had a slightly higher SSIM gain (`+0.01161`) but lower PSNR and fewer wins, so step 39000 is the more balanced public default.

The continuation improved mean PSNR on every tested degradation. However, the strong `photo_v2` and `photo_v3_noise_mix` win counts fell versus step 11000, indicating that the larger correction has a less conservative failure tail. A post-training residual-strength guardrail provides an explicit trade-off: strength `1.0` gives the best average quality, `0.75` preserves most of the gain with fewer regressions, and `0.5` raises strong-preset wins from `81/100` to `86/100` while keeping positive mean gains. These modes are exposed in the inference CLI and Colab. Future work should prioritize user-facing evaluation and a learned degradation-aware gate rather than continuing indefinitely.

18 Systems Notes

Diffusion training now supports PyTorch DDP when launched with `torchrun`. Without `torchrun`, the same script falls back to the existing single-GPU path. On the tested 2x A100 SXM environment, a 1 GiB NCCL all-reduce smoke test reported about `199.5 GB/s`, so multi-GPU communication was not the bottleneck.

The completed Stage 4 XL edge run trained at about `0.78 step/s` in ordinary train sections. A 5000-step run is roughly 1.8 hours of pure training, plus eval/checkpoint overhead.

On the tested 2x L40S environment, the gated-residual probe ran with high GPU utilization, about `0.79` micro-step/s, and roughly `44.7 / 46.1 GiB` allocated per GPU. No GPU bottleneck or NCCL communication issue was observed.

The decoded-detail residual refiner v2 ran on one L40S at about `0.89` micro-step/s with `42.0 / 46.1 GiB` allocated and sustained `99-100%` GPU utilization.

The guarded-detail Stage 2 v2 continuation from dual-context LSDIR step 98000 was stopped after 20000 micro-steps because the run had plateaued. Its selected checkpoint is the step-10000 `best_eval_mean_psnr_detail.pt`, not the final step. This candidate is much lighter at inference time because it only runs the Stage 2 encoder and Stage 1 decoder: the Stage 2 model has `119.24M` parameters, the training checkpoint is about `1.4GB`, and the model weights inside it are about `0.44GB`. On one L40S with bf16 and 128x128 LR tiles, measured CUDA reserved memory was `1.03GB` at tile batch 1, `3.28GB` at tile batch 4, and `6.25GB` at tile batch 8. Practical inference can therefore run on an 8GB GPU with tile batch 1, while 12-16GB GPUs are comfortable for local review. Long Stage 2 training still targets a 48GB GPU class machine.

19 Visual Review and External Positioning

A fixed-sample visual review now compares LR, bicubic, Stage 2 condition, residual strengths `0.50`, `0.75`, and `1.00`, and GT side by side. On mild inputs the selected refiner makes small, generally structure-preserving improvements. On stronger `photo_v2` and `photo_v3_noise_mix` inputs, the main failure occurs earlier: the condition encoder removes noise together with real detail and sometimes leaves small cyan/white grid-like artifacts. The residual refiner changes are too small to recover the missing fur, leaf veins, text, and distant structural detail.

Qualitatively, the current model is not competitive with leading generative restoration systems in perceived sharpness or plausible fine texture. Its useful distinction is a deterministic, vision-only path trained without a pretrained text-to-image model, with lower hallucination risk and adjustable correction strength. This is an architectural and visual assessment, not a direct SOTA benchmark. A defensible comparison requires same-input blind A/B testing plus LPIPS/DISTS/MANIQA/MUSIQ and user-preference evaluation.

19.1 Representative Visual Examples

The current README demo alternates `Set5/img_003` between noisy LR input and LuSIR output. A fixed Gaussian degradation (`sigma=4/255`, seed `20260622`) is used; the noisy bicubic baseline is omitted from the animation for clarity but is retained for metric calculation. Masked detail v2 improves over that noisy bicubic baseline by `+5.17 dB` Y PSNR and `+0.163` Y SSIM. Because the PDF cannot animate that comparison, the static report figure below uses `Urban100/img_043`, the strongest measured clean-bicubic full-image example from the fixed x4 benchmark among the reviewed candidates. It is intentionally used as a high-impact visual explanation rather than as a claim of average SOTA quality: masked detail v2 improves this sample over bicubic by `+10.66 dB` Y PSNR and `+0.152` Y SSIM, mainly by restoring repeated architectural structure.

Figure 1. High-impact full-image benchmark example. The method still remains a conservative deterministic restoration path; this image is chosen because the improvement is easy to see.

The following two lime examples use the same validation crop so that the role of the base reconstruction and the later detail branch can be inspected separately. They are diagnostic examples, not claims of a formal benchmark win.

Figure 2. Multiscale Stage 2 restores the degraded input while retaining a visible fine-texture gap to ground truth.

Figure 3. Selected detail branch v1d adds a controlled, artifact-light correction over the base; this illustrates both its improvement and its remaining conservative behavior.

20 Stage 2 Multiscale Redesign

A decoded pixel/edge/highpass fine-tuning probe confirmed that the flat Stage 2 condition predictor is the current bottleneck. By step 4000, decoded PSNR had risen from `23.7387` to `24.2792`, but the Laplacian detail ratio fell from `0.2817` to `0.2773` and fixed samples remained visibly smooth. This is consistent with the distortion-perception limitation of optimizing reconstruction losses without sufficient spatial context or high-quality training exposure.

The replacement keeps all parameters and names of the selected 19M-parameter step 72000 predictor, then adds a zero-output-initialized 128-to-64-to-32 multiscale context branch. Partial initialization therefore exactly preserves the previous output before training. The training manifest also changes from 100,000 COCO plus 3,450 DIV2K/Flickr2K rows to a roughly balanced exposure of 100,000 COCO and 103,500 repeated DIV2K/Flickr2K rows. Random crops and degradations remain stochastic, so repeated high-quality rows generate different training pairs.

The design is informed by broad-context and multiscale restoration findings in SwinIR, HAT, and NAFNet, together with the data/degradation emphasis of Real-ESRGAN. The resulting 55.50M-parameter model passed a full batch-8 forward/backward and val100 smoke test on one L40S at approximately `34.8/46.1GB` VRAM and `100%` GPU utilization. The 50,000-micro-step long run is tracked at https://wandb.ai/jwheo/LuSIR/runs/6zt2do4v.

The run completed normally. Step 46000 was selected over the final checkpoint because it provides the best cross-preset distortion/detail compromise. It improves the previous Stage 2 by `+1.0348 dB` on `photo_detail_mix` and `+0.9228 dB` on `mild`, with `99/100` and `97/100` per-image wins and slightly higher detail ratios. On the stronger `photo_v2` and `photo_v3_noise_mix` presets it improves PSNR by `+0.9441 dB` and `+0.9650 dB`, but the detail ratio drops from roughly `0.29-0.30` to `0.22-0.23`. Visual comparison confirms that the model improves color, large boundaries, and denoising without convincingly recovering missing fur, text, fabric, or distant texture. The experiment is a successful base-reconstruction redesign but not a solution to perceptual fine-detail recovery.

An optional continuation completed from step 46000 using frozen ImageNet VGG16 features at shallow and intermediate layers. This explicitly introduces pretrained vision feature supervision, while still avoiding pretrained text-to-image or generative models. A CUDA smoke test passed at batch 4 with approximately `20.6/46.1GB` VRAM and `2.62` micro-steps/s. The run completed 12,000 micro-steps. Step 8000 had the best shortlist score, `26.0092`, and improved decoded PSNR over initialization by `+0.0101` to `+0.0256 dB` across the four tested presets. Step 11000 was strongest on cleaner presets but regressed `photo_v3_noise_mix` by `-0.0063 dB`. Fixed contact sheets showed almost no visible difference from initialization, and missing fine texture remained smoothed. The run is preserved as a partial metric/latent success but is not promoted into the public path. The completed run is tracked at https://wandb.ai/jwheo/LuSIR/runs/nrqhw05u.

21 Dual-Context Unique-Data Stage 2 Scale-up

The VGG continuation's `+0.0101` to `+0.0256 dB` cross-preset improvements were not visually distinguishable and did not recover missing texture. Capacity and data uniqueness are therefore changed together instead of extending the same objective again. The previous HQ-balanced manifest had 203,600 rows but only 103,550 unique images because DIV2K and Flickr2K were repeatedly exposed.

The new manifest adds 30,000 unique LSDIR images for 133,450 unique training images plus the unchanged 100-image validation set. The predictor retains the selected 55.50M-parameter multiscale model and adds a second zero-output- initialized multiscale context branch. Partial initialization exactly preserves the selected step 46000 output before training while increasing capacity to 119.24M parameters.

The one-L40S smoke test reproduced the initial `24.48 dB` decoded PSNR and `0.291` detail ratio. Batch 8 with gradient accumulation 4 used approximately `37.8/46.1GB` VRAM, reached `99%` GPU utilization, and sustained approximately `0.75` micro-step/s. The completed long run targeted 100,000 micro-steps, or 25,000 optimizer updates. Evaluation ran every 1,000 micro-steps, while ordinary milestone checkpoints were limited to every 5,000 micro-steps to stay within the disk budget. The run is tracked at https://wandb.ai/jwheo/LuSIR/runs/4akqckxu.

The run completed all 100,000 micro-steps. Step 98,000 is the automatic `eval/decoded_psnr` best checkpoint on `photo_detail_mix`; step 100,000 is slightly better on stronger degradations. Re-evaluated with the same comparison tool against selected step 46,000, the gains are `+0.1362 dB` on `photo_detail_mix`, `+0.1086 dB` on `mild`, `+0.0540 dB` on `photo_v2`, and `-0.0356 dB` on `photo_v3_noise_mix` for the best checkpoint. The final checkpoint changes those to `+0.1256`, `+0.1025`, `+0.0668`, and `+0.0132 dB`. This is a modest reconstruction improvement, not a solved perceptual-detail problem.

22 High-Frequency Detail Branch v1

The latest deterministic detail candidate is an image-space detail branch, not another Stage 4 continuation. The path is:

```
LR -> frozen Stage 2 dual-context condition -> frozen Stage 1 decoder -> base SR
base SR + bicubic LR upsample -> gated high-frequency detail branch -> detail SR
```

The branch predicts a bounded RGB residual and a per-pixel gate. The residual is projected through a local highpass operation before being added to the base SR, which limits the model's ability to change global color or low-frequency structure. The output convolution is zero-initialized, so step 0 exactly reproduces the frozen Stage 2 + Stage 1 base output.

Implemented files:

```
configs/detail_branch_v1_photo130k_lsdir.yaml
configs/detail_branch_v1b_aug_photo130k_lsdir.yaml
configs/detail_branch_v1c_condition_open_photo130k_lsdir.yaml
configs/detail_branch_v1d_deep3m_photo130k_lsdir_3ep.yaml
configs/hf/detail_branch_v1d_deep3m_photo130k_lsdir_3ep.yaml
```

```
tools/train/train_detail_branch.py
tools/eval/run_fixed_review_detail_branch.py
tests/test_detail_branch.py
```

A four-micro-step smoke test completed load/eval/backprop/update/checkpoint successfully. Step 0 matched the base output exactly at `24.6188 dB` val100 PSNR; after one optimizer update, the branch produced a tiny `+0.00005 dB` aggregate PSNR delta and `69/100` wins versus base. This is only an integration check, not a quality claim.

The first v1 run was stopped at 7800 micro-steps, or 0.234 epoch over the 133450-image train manifest, because early samples showed that the residual was safe but visually too small. The v1b run kept the same model and loss, but added horizontal flips, texture-biased crop retry, and weak HR color jitter while excluding rotation, vertical flip, affine/perspective transforms, erasing, and mixup.

The v1b run completed 40000 micro-steps, equal to 10000 optimizer updates with `grad_accum_steps: 4`. It selected step 39500 by `eval/detail_score`:

```
base PSNR:        24.6188
detail PSNR:      24.6649
PSNR delta:       +0.0461 dB
base SSIM:        0.80013
detail SSIM:      0.80281
SSIM delta:       +0.00268
mean PSNR delta:  +0.0575
wins vs base:     98/100
detail wins:      100/100
```

Different metrics peak at nearby checkpoints: PSNR delta is highest at step 38500 (`+0.0489 dB`), SSIM delta is highest at step 37000 (`+0.00336`), and the final step 40000 reaches `+0.0444 dB` PSNR and `+0.00277` SSIM with `98/100` wins. The selected step 39500 remains preserved as the earlier public detail artifact because it has the best combined detail score from the completed v1b run. Qualitatively, it is artifact-light and slightly sharper on texture-heavy crops, but still conservative and below GT on fine surface detail. It remains a historical comparison artifact; the later selected v1d checkpoint is exposed through the Colab WebUI detail runner.

V1c initialized from the selected v1b checkpoint, exposed the frozen Stage 2 condition latent directly to the image-space branch, increased the bounded residual scale, and opened the gate slightly. The selected step 6000 improves the fixed `photo_detail_mix` val100 base by `+0.0554 dB` aggregate PSNR and `+0.00332` SSIM with `99/100` wins. This was better than v1b but plateaued quickly enough to motivate a controlled capacity test.

V1d keeps the v1c width and objective but increases residual depth from 8 to 18 blocks, raising branch capacity from `1.35M` to `3.02M` parameters. The original blocks are copied from v1c step 6000 and the ten appended blocks are identity-initialized, so the step-0 output exactly reproduces v1c. The run completed 100,086 micro-steps, exactly three passes over the 133,450-image manifest at batch size 4, and selected step 99,500 by `eval/detail_score`.

On ordinary fixed `photo_detail_mix` val100, selected step 99,500 improves the frozen base by `+0.1646 dB` aggregate PSNR, `+0.1888 dB` mean PSNR, and `+0.00647` SSIM with `99/100` PSNR wins and `100/100` detail wins. On the strict-bicubic five-crop diagnostic it reaches `31.9513 dB`, improving the frozen base by `+0.2102 dB`, v1c by `+0.1358 dB`, and winning `5/5` images. This is also `+0.1266 dB` above the early v1d step-9500 snapshot.

Final step 100,086 reaches a nearly identical strict-bicubic `31.9516 dB`, but selected step 99,500 is stronger on ordinary-val aggregate PSNR, SSIM, highpass improvement, and the combined detail score. Visual inspection shows no white-dot, grid, or excessive-sharpening artifacts. The larger branch and long run therefore produced meaningful stable progress, but still did not close the visible fine-texture gap to ground truth. Further same-objective continuation or simple capacity scaling is not planned.

23 Learned Detail-Need Mask and Masked Detail Branch v2

The next experiment separated `where detail is missing` from `what detail should be generated`. A compact 460,545-parameter predictor observes the frozen base reconstruction, bicubic input, Stage-2 condition latent, and four inference-time detail proxies. On fixed `photo_detail_mix` val100, predictor step 3250 raises pixel correlation with the GT-supervised detail-need target from `0.5403` to `0.7456`, raises top-20% missing-detail capture from `0.3252` to `0.3861`, and lowers excess-detail capture from `0.4838` to `0.4304`.

Masked detail branch v2 initializes from v1d step 99500 and applies the frozen predictor as a soft spatial gate with a `0.05` floor. Two independent continuations converged to nearly identical candidates:

Checkpoint	Detail score	PSNR delta	Mean PSNR delta	SSIM delta
step 36000	26.69528	+0.18744 dB	+0.20521 dB	+0.00721
selected step 38000	**26.69601**	+0.18177 dB	+0.20432 dB	**+0.00755**

Step 38000 is selected by the configured `eval/detail_score`. No new best appeared through step 50000, and fixed grids at steps 34000, 38000, and 48000 were nearly indistinguishable, so the run stopped before its 20-epoch upper bound.

The predictor establishes that missing-detail localization is learnable from inference-time observations. However, gating the same deterministic L1/highpass-oriented branch does not visibly synthesize the missing fine texture. Location selection and texture generation remain separate problems. The first top-fraction texture-gate review also showed why the mask objective needs explicit negative examples. With a top-10% binary gate, v1 selected useful texture-rich regions on clean inputs, but when synthetic Gaussian noise was injected into a low-texture patch, `45.31%` of that patch fell inside the top-10% gate and the predictor mean inside the noisy patch rose to `0.6430`. The GT-supervised missing-detail target stayed low there, so this was not desired texture discovery; it was the predictor opening on artifact-like high-frequency content.

A noise-negative v2 mask probe therefore initializes from v1 step 3250 and adds low-target patch noise augmentation plus explicit in-patch prediction and excess penalties. Its selected step 1500 preserves clean top-10 selection quality (`0.7219` score versus `0.7173` for v1, with excess capture improving from `0.2496` to `0.2375`) while reducing the same injected-noise top-10 coverage from `0.4531` to `0.0000` and noisy-patch predictor mean from `0.6430` to `0.0018`. This makes v2 the preferred gate for any subsequent texture-generation branch.

The follow-up v5 texture probe used this v2 mask as a hard top-10% binary gate with zero floor and retried masked VGG plus PatchGAN pressure from the selected v2 generator. The gate itself behaved as intended: `detail_mask_mean` stayed near `0.10` and `outside_mask_residual_l1` remained `0`. However, the adversarial phase still collapsed fidelity inside the mask. The run moved from `+0.0537 dB` PSNR delta and `99/100` wins at step 500 to `-0.0953 dB` and `11/100` wins at step 3500. Visual review showed high-frequency artifact growth rather than GT-aligned fine texture. This rejects the current PatchGAN route; stronger adversarial pressure or longer continuation is not planned.

The implemented v6 follow-up retains the frozen fidelity base and v2 noise-negative top-10 mask, but removes the PatchGAN objective. It starts from selected masked detail v2 step 38000 and combines masked VGG supervision, GT-filtered RealESRGAN highpass/residual teacher signal, and a new artifact-negative residual loss. The negative loss directly suppresses residual highpass energy in flat target regions or regions where the base is already sharper than the target. Its config is `configs/detail_branch_v6_noise_gate_teacher_perceptual_no_gan_probe.yaml`, with design and stop criteria in `docs/DETAIL_BRANCH_V6_NO_GAN_KO.md`.

A four-step smoke run confirmed that the v6 wiring preserves the intended guardrails before long training: `eval/detail_mask_mean` is `0.1000`, `eval/outside_mask_residual_l1` is `0.000000`, `eval/lowpass_drift_l1` is `0.000130`, and the initialized path remains positive versus the frozen base with `+0.0696 dB` mean PSNR and `+0.00147` SSIM. These values are not a promotion result; the actual decision depends on 500-6000 step W&B sample grids and whether the new supervision creates real texture without v5-style scratch/noise artifacts.

The completed 6000-step v6 run did not collapse, but it also did not improve over its initialization. The selected checkpoint is step 0. Final step 6000 still keeps `eval/outside_mask_residual_l1` at `0.000000` and remains positive versus the frozen base (`+0.0534 dB` mean PSNR and `+0.00103` SSIM), but it is below the initialized score and the fixed grids are visually almost unchanged. This is useful negative evidence: no-GAN teacher/negative losses can keep the branch safe, but this image-space branch still does not create the missing texture.

The follow-up Stage 2 latent residual adapter v1 moved the residual correction into latent space. `configs/latent_pretrain_photo` freezes the dual-context Stage 2 best98000 predictor, feeds its latent output and LR input into a zero-initialized 3.75M-parameter adapter, and trains only a bounded latent residual before decoding through the same Stage 1 decoder. Its starting output is exactly the frozen Stage 2 base, so any improvement or regression can be attributed to the adapter rather than a full Stage 2 retraining drift.

The run completed 12000 micro-steps and selected step 11000 on the val100 composite metric. It was stable, but not useful enough to promote. On the formal 219-image benchmark, it is slightly worse than the frozen Stage 2 base in mean Y PSNR (`27.8294` versus `27.8431`) and lower than the guarded Stage 2 v2 default in both Y PSNR and Y SSIM. This rules out plain latent residual adapter continuation as the next priority.

Before v5, the v3/v3b/v4 series tested the same general idea with the original v1 mask. V3 added conservative masked VGG and PatchGAN losses and selected step 1000, but remained visually too close to the deterministic v2 branch. V3b increased perceptual and adversarial pressure; its final step 8000 fell to `-0.1824 dB` PSNR delta and `11/100` wins. V4 added filtered Real-ESRGAN teacher highpass supervision, but its best checkpoint was step 0, meaning the teacher signal did not improve over the v3 initialization. V5 shows that switching to the cleaner v2 noise-negative gate fixes mask leakage but does not fix the adversarial texture objective itself.

The evaluator reports lowpass drift and residual energy inside and outside the learned mask. These values, PSNR/SSIM, wins, and fixed visual grids remain the guardrails for any future detail-generation probe.

During reproducibility review, the original single-image and formal-benchmark inference paths were found to omit the learned predictor even though training and validation applied it. The inference and benchmark runners now load the predictor checkpoint and floor from the HF config. V1d configs without a mask retain their previous behavior.

The masked v2 checkpoint also completed the formal 219-image clean-bicubic benchmark. Relative to v1d, Y PSNR improves by `+0.0034`, `+0.0602`, `+0.0135`, and `+0.0167 dB` on DIV2K, Set5, Set14, and Urban100. The overall gain is `+0.0114 dB` Y PSNR and `+0.00118` Y SSIM. This consistent but small gain supports the same conclusion as the fixed grids: learned gating improves correction placement, but does not solve visible texture synthesis.

24 Public Artifacts

The latest LuSIR public artifacts are stored in `jwheo/LuSIR` on Hugging Face:

```
checkpoints/stage4_photo100k_xl_edge_b16_best_eval_condition_decoded.pt
checkpoints/residual_refiner_stage2_xl_mild_best_eval_refined.pt
checkpoints/stage4_photo100k_xl_teacher_residual_photo_v3_step_0002000.pt
checkpoints/stage4_photo100k_xl_teacher_residual_photo_detail_best8000.pt
checkpoints/residual_refiner_stage2_xl_photo_detail_v2_best39000.pt
checkpoints/stage2_photo100k_multiscale_hqmix_step_0046000.pt
checkpoints/stage2_photo100k_multiscale_hqmix_perceptual_step_0008000.pt
checkpoints/stage2_photo130k_lsdir_dual_multiscale_best98000.pt
checkpoints/detail_branch_v1b_aug_photo130k_lsdir_best39500.pt
checkpoints/detail_branch_v1d_deep3m_photo130k_lsdir_best99500.pt
checkpoints/detail_mask_predictor_v1_best3250.pt
checkpoints/detail_branch_v2_masked_photo130k_lsdir_best38000.pt
metrics/stage4_photo100k_xl_edge_b16_val100_t50_32step_summary.json
metrics/diagnose_stage2_xl_residuals_mild_val100_summary.json
metrics/residual_refiner_stage2_xl_mild_probe_early_stop_summary.json
metrics/eval_residual_refiner_stage2_xl_photo_v3_noise_mix_val100_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t25_32step_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t50_32step_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_detail_best8000_val100_t25_summary.json
metrics/residual_refiner_stage2_xl_photo_detail_v2_long_summary.json
metrics/residual_refiner_v2_best39000_strength_sweep_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_mild_val100_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_photo_v2_val100_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_photo_v3_noise_mix_val100_summary.json
metrics/stage2_multiscale_hqmix_step46000_cross_preset_summary.json
metrics/stage2_multiscale_perceptual_photo_detail_mix_candidates.json
metrics/stage2_multiscale_perceptual_mild_candidates.json
metrics/stage2_multiscale_perceptual_photo_v2_candidates.json
```

```
metrics/stage2_multiscale_perceptual_photo_v3_noise_mix_candidates.json
metrics/stage2_photo130k_lsdir_dual_multiscale_final_summary.json
metrics/detail_branch_v1b_aug_photo130k_lsdir_summary.json
metrics/detail_branch_v1d_deep3m_photo130k_lsdir_3ep_summary.json
metrics/benchmark_bicubic5_detail_v1d_best99500_summary.json
metrics/detail_branch_v2_masked_photo130k_lsdir_summary.json
metrics/formal_x4_benchmark_detail_v2_masked_summary.json
metrics/formal_x4_benchmark_detail_v2_masked_metrics.csv
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_summary.json
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_metrics.csv
metrics/formal_x4_benchmark_stage2_detail_perceptual_v1_delta_crop_selection.csv
metrics/formal_x4_benchmark_stage2_latent_adapter_v1_value_compare_summary.json
metrics/formal_x4_benchmark_stage2_latent_adapter_v1_value_compare_metrics.csv
samples/stage4_photo100k_xl_edge_b16_val100_t50_32step_grid_lr_bicubic_sr_gt.png
samples/diagnose_stage2_xl_residuals_mild_val100_grid.png
samples/residual_refiner_stage2_xl_mild_probe_step500_grid.png
samples/eval_residual_refiner_stage2_xl_photo_v3_noise_mix_val100_grid.png
samples/compare_residual_refiner_vs_stage4_edge_0801_photo_v3.png
samples/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t25_grid.png
samples/stage4_photo100k_xl_teacher_residual_photo_detail_best8000_val100_t25_grid.png
samples/residual_refiner_stage2_xl_photo_detail_v2_best39000_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_mild_val100_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_photo_v2_val100_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_photo_v3_noise_mix_val100_grid.png
samples/stage2_multiscale_hqmix_checkpoint_comparison.png
samples/stage2_multiscale_perceptual_photo_detail_mix_candidates.png
samples/stage2_multiscale_perceptual_photo_v3_noise_mix_candidates.png
samples/stage2_dual_lsdir_photo_detail_mix_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_mild_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_photo_v2_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_photo_v3_noise_mix_best98k_final100k_contact_sheet.png
samples/detail_branch_v1b_aug_photo130k_lsdir_best39500_grid.png
samples/detail_branch_v1d_deep3m_photo130k_lsdir_best99500_grid.png
samples/benchmark_bicubic5_detail_v1d_best99500_grid.png
samples/detail_branch_v2_masked_photo130k_lsdir_best38000_grid.png
samples/stage2_detail_perceptual_v1_benchmark_delta_crop_sheet.jpg
samples/stage2_latent_adapter_v1_value_compare_selected.jpg
samples/stage2_latent_adapter_v1_value_compare_contact_sheet.jpg
docs/assets/lusir_current_demo_urban100_img043.jpg
docs/assets/lusir_current_demo_set5_butterfly.gif
configs/diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml
configs/residual_refiner_stage2_xl_mild_probe.yaml
configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_v3_b8_probe.yaml
configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_detail_b8_long.yaml
configs/residual_refiner_stage2_xl_photo_detail_v2_continue_40k.yaml
configs/hf/residual_refiner_stage2_xl_photo_detail_v2.yaml
configs/latent_pretrain_photo100k_multiscale_hqmix_perceptual_continue.yaml
configs/latent_pretrain_photo130k_lsdir_dual_multiscale_long.yaml
configs/detail_branch_v1_photo130k_lsdir.yaml
configs/detail_branch_v1b_aug_photo130k_lsdir.yaml
configs/hf/detail_branch_v1b_aug_photo130k_lsdir.yaml
configs/detail_branch_v1d_deep3m_photo130k_lsdir_3ep.yaml
configs/hf/detail_branch_v1d_deep3m_photo130k_lsdir_3ep.yaml
configs/hf/detail_mask_predictor_v1.yaml
configs/hf/detail_branch_v2_masked_photo130k_lsdir.yaml
configs/latent_pretrain_photo130k_lsdir_latent_residual_adapter_v1.yaml
```

25 Next Work

The guarded-detail Stage 2 v2 step-10000 checkpoint is now the public Colab default because it is the best current T4-friendly deterministic path. Residual refiner v2 remains the conservative deterministic option, while detail branch v1d and masked detail branch v2 are selectable single-image/tiled Colab research options. The completed perceptual Stage 2 continuation is not promoted. Candidate next steps are:

- do not continue masked latent residual-shift in its current form. The v1 full-trajectory best at step 3500 increased detail ratio from 0.7965 to 0.8272 but lost 0.7557 dB PSNR and worsened GT-aligned high-frequency error. Correction-strength and start-timestep sweeps showed that settings preserving fidelity also removed nearly all visible effect. The v2 fidelity continuation reduced the loss to 0.0651 dB PSNR and 0.00066 SSIM below base, but detail ratio increased only to 0.8003 and high-frequency error remained slightly worse. The branch is therefore not promoted;
- do not continue the v6/v7/v8 detail-branch objectives without a new signal. V6 was safe but non-improving, v7 showed that RealESRGAN teacher patches were not reliable under the GT highpass filter, and v8 showed that training-time GT detail masks can preserve a small stable correction but do not create a visible texture breakthrough. V8 selected step 500; step2000 was stopped after laplacian ratio turned slightly negative;
- do not continue the Stage 2 latent residual adapter v1 objective without a new supervision signal; it completed stably but trailed guarded Stage 2 v2 on the formal benchmark;
- preserve the deterministic base and move from full latent re-prediction to a residual/detail correction path that predicts `target_latent - baseline_latent` or decoded highpass/detail residual over the existing Stage 2 output;
- use the completed Stage 1 audit result as a guardrail: the decoder preserves high-frequency detail when fed HR latents, so the next bottleneck to attack is Stage 2 conditional-mean smoothing rather than decoder capacity;
- do not continue `configs/latent_pretrain_photo130k_lsdir_dual_detail_gtmasked_v3_probe.yaml` in its current form. The train-only `prediction_missing` top20 masked decoded/highpass objective increased mean PSNR slightly by step1000, but highpass ratio fell from `0.8084` to `0.794` and missing-detail energy rose from `0.01897` to `0.01939`. The next Stage 2 attempt needs a changed target parameterization or architecture, not another mask-weighted loss continuation;
- use `configs/latent_pretrain_photo130k_lsdir_dual_bicubic_overfit64_probe.yaml` only as a non-deployable upper-bound check. It completed successfully: train64 mean PSNR rose from `26.4246` to `29.2477`, highpass ratio from `0.7908` to `0.8812`, and missing energy from `0.01971` to `0.01314`. Therefore the next Stage 2 work should focus on data/curriculum/regularization that generalizes this detail signal, not another fixed-subset continuation;
- select the generative path with LPIPS, DISTS, fixed visual review, high-frequency metrics, lowpass drift, and seed diversity instead of PSNR alone;
- keep Stage 2/base architecture research separate from the generative detail objective, but do not continue the shifted-window attention/window-scaling branch without a new supervision signal;
- do not spend the next iteration on larger TTA: full x8 improves guarded Stage 2 by only `+0.0957 dB` mean Y PSNR while preserving the same visibly conservative texture character;
- because v1d remains visually conservative, noise-MSE residual diffusion collapses toward zero, and v5 PatchGAN collapsed inside the mask, prioritize patch-level perceptual and artifact-negative supervision before trying adversarial pressure again;
- keep a degradation-aware gate or strong-input guardrail as the primary response to the remaining strong-preset failure tail.

The first latent residual probe increased high-frequency energy but worsened GT-aligned detail and was stopped. The replacement diffuses only signed Haar high bands of the image-space residual over the frozen v1d base. Its clipped oracle improves the val8 base from `28.7012` to `31.4039` dB, confirming that the representation can express useful corrections. The step-20,000 condition-start continuation removed the early stochastic grain, but the residual magnitude and seed diversity also collapsed toward zero. On val100, start timesteps 15, 25, and 50 trail the v1d base by `0.0880`, `0.1392`, and `0.3152` dB respectively, and all settings worsen GT-aligned Laplacian and highpass error. The result is not promoted, and the same noise-MSE objective will not be continued. The next generative detail experiment should use an explicit learned detail-need mask plus patch-level perceptual or adversarial supervision rather than relying on conditional-mean residual prediction. Implementation and evaluation details are in `docs/HIGH_FREQUENCY_RESIDUAL_DIFFUSION_KO.md`.

The learned-mask prerequisite was subsequently trained as a compact 460,545-parameter predictor using the frozen fidelity base, bicubic image, Stage-2 condition latent, and four observable detail proxies. On the fixed photo-detail val100 set, it improved pixel correlation with the supervised detail-need target from `0.5403` to `0.7456`, improved top-20% missing-detail capture from `0.3252` to `0.3861`, and reduced excess-detail capture from `0.4838` to `0.4304`. This clears the predefined localization gate, but does not by itself establish improved SR output. The subsequent masked v2 branch modestly improved ordinary val100 metrics but remained visually close to v1d and plateaued by step 38000. Therefore the same deterministic objective will not be continued; the next experiment must change texture-generation supervision while keeping the validated spatial gate and fidelity guardrails.